

AMGF-Schwarz

Improving the Scalability of Algebraic Multigrid with Filtering
Summer 2026

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Center for Applied Scientific Computing (CASC)

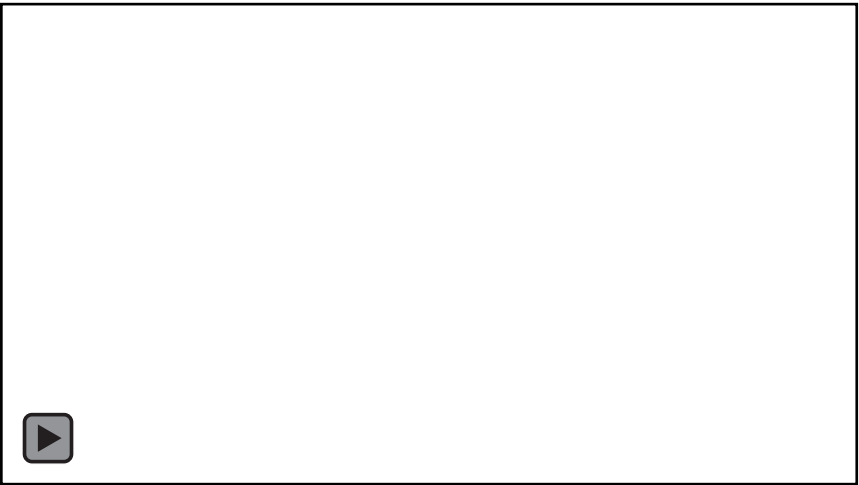
Prepared by LLNL under Contract DE-AC52-07NA27344.



Background

Modeling many physical systems requires contact

Cloth



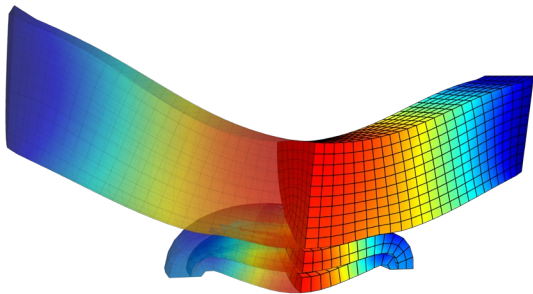
Huang et al. 2025

Robotics



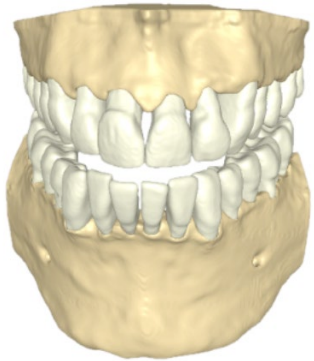
Huang et al. 2025

Engineering



Petrides et al. 2026

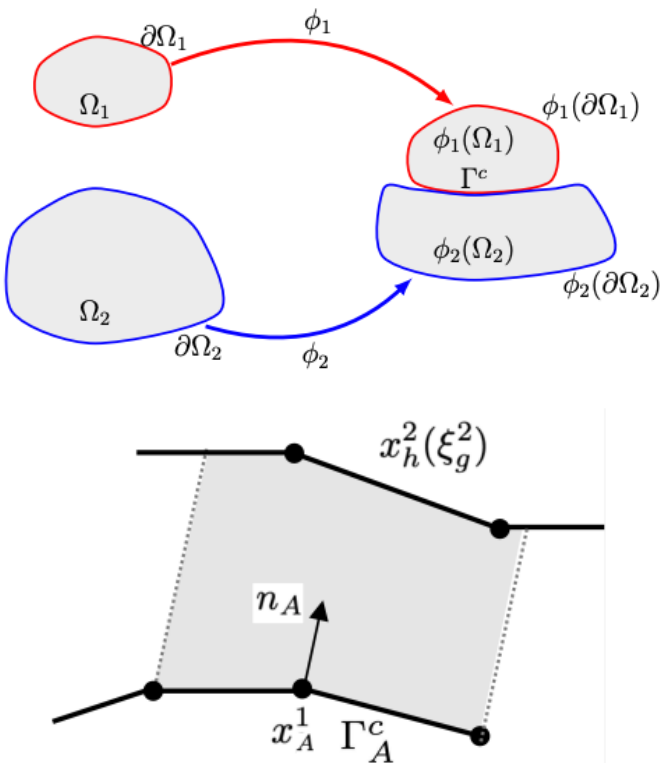
Biomechanics



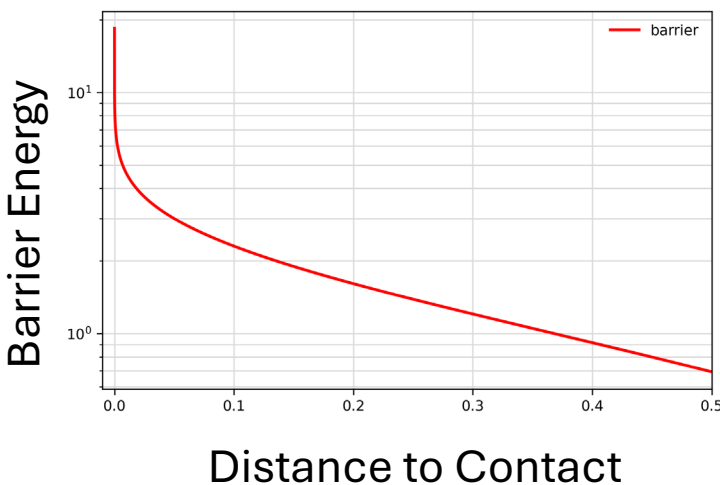
Gholamalizadeh et al. 2022

Goal: scalable and robust numerical methods for contact problems.

We solve an inequality-constrained optimization problem



- The interior-point iterations require solving ill-conditioned linear systems.



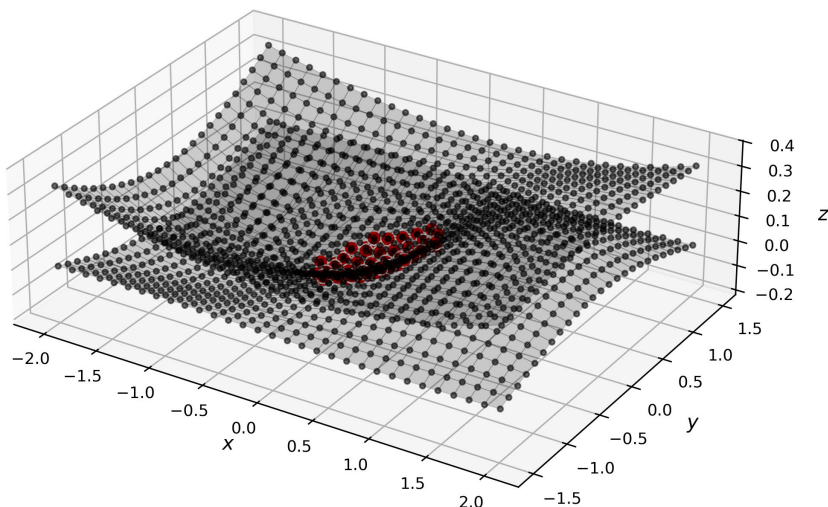
$$A = K + J^T D J$$

K : stiffness matrix w/o contact, J : Jacobian of constraint, D : log-barrier Hessian

Puso and Larsen 2004

Algebraic Multigrid with Filtering (AMGF)

- AMGF applies AMG V-cycles sandwiched around an exact subspace correction via direct factorization.
- The increase in PCG iterations due to contact is bounded by a factor of about $\sqrt{2}$.



Error Reduction Matrix from Petrides et al. 2026

$$I - MA = (I - BA)(I - PA_c P^T A)(I - BA)$$

M : AMGF preconditioner, B : global AMG V-cycle, P : subspace selection matrix, A_c : contact submatrix

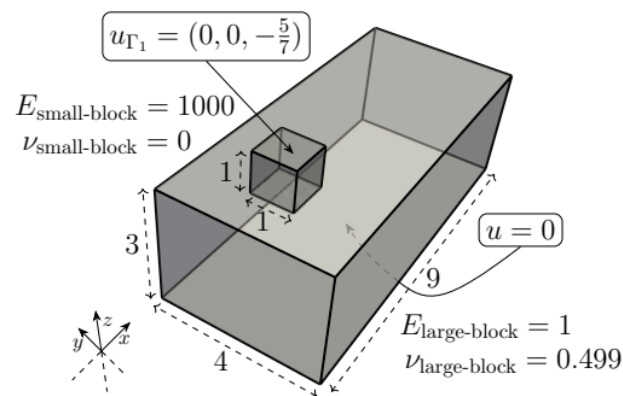
For large, contact-rich problems, direct factorization of the subspace will become a bottleneck

	Direct	Iterative
Time Scaling	$O(n^2)$	$O(n)^*$
Memory Scaling	$O(n^{\frac{4}{3}})$	$O(n)$
Sensitivity to ill-conditioning	Largely insensitive.	Convergence degrades without specialized preconditioners.

A more scalable alternative is needed.

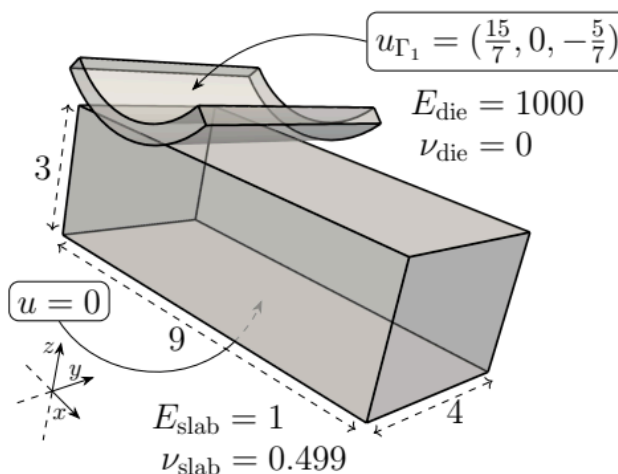
We consider three model problems

Two-block

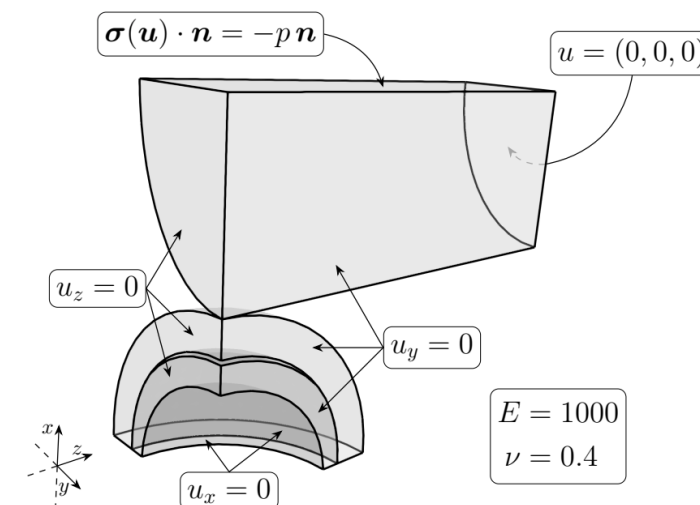


(a) Initial configuration

Ironing



Beam-sphere



We implement and test AMGF-Schwarz in MFEM using extended versions of the *AMGFSolver* class and contact miniapp.

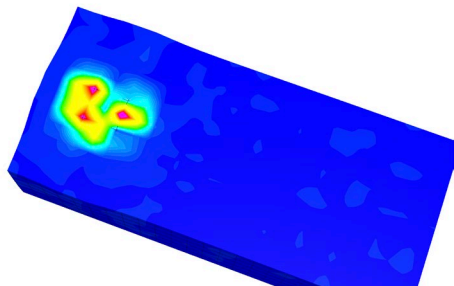


Method

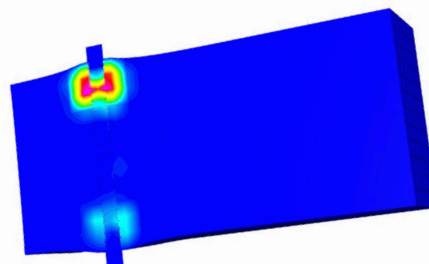
The difficult eigenmodes have local structure

- The log-barrier term introduces large eigenvalues, degrading system conditioning.
- We note that the corresponding eigenvectors are highly localized.

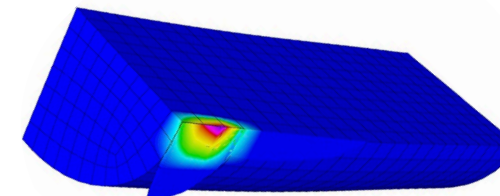
Two-block



Ironing

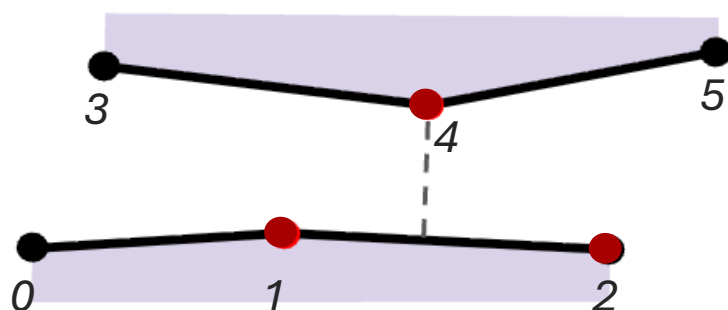


Beam-sphere



AMGF-Schwarz performs an overlapping additive Schwarz correction based on individual constraints

- w accounts for overlap, which we set empirically to 0.1.



$$\begin{array}{cccccc} & \text{First row of } J & & & & \text{First Schwarz subdomain} \\ (0 & j_1 & j_2 & 0 & j_4 & 0) \rightarrow \mathcal{J}_1 = \{1, 2, 4\} \end{array}$$

$$R_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

AMGF-Schwarz error reduction matrix

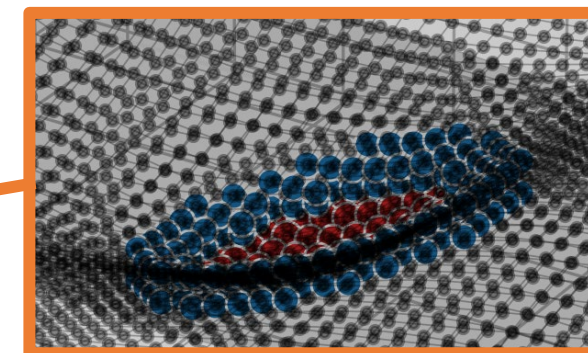
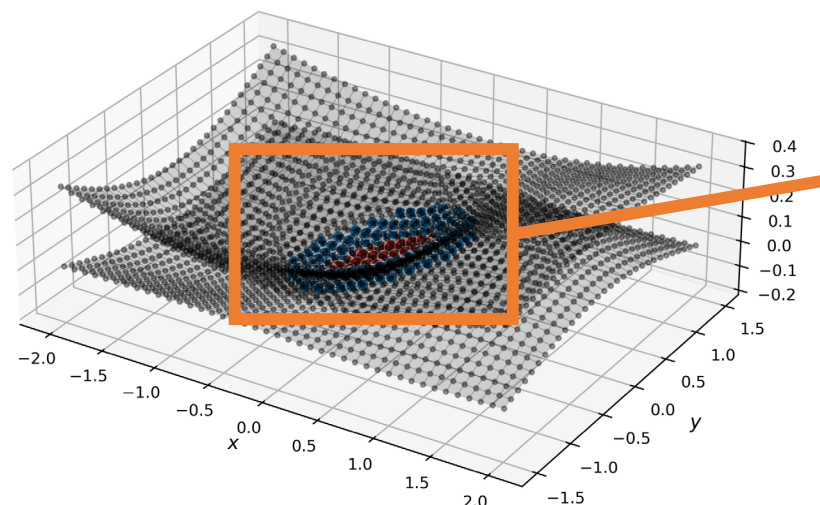
$$I - MA = (I - BA)(I - w \sum_k R_k^T (R_k A R_k^T)^{-1} R_k A)(I - BA)$$

Not all constraints are equally difficult

- Only nearly violated constraints introduce massive eigenvalues.
- We can skip the subspace correction for constraints that are not near violation, which corresponds with small entries in D .

$$D \sim \frac{1}{g^2}$$

g : gap function



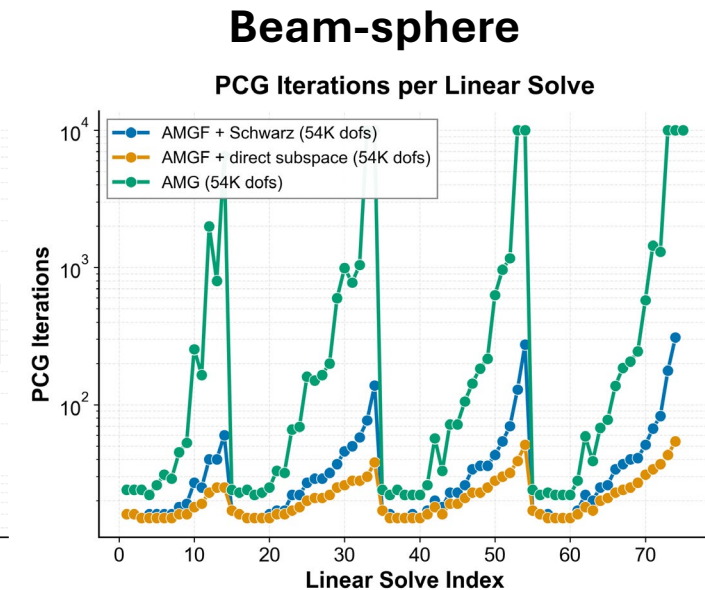
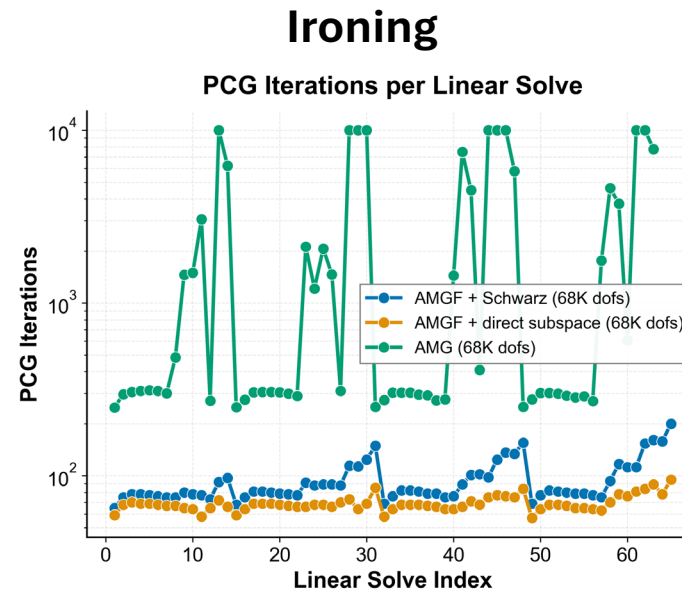
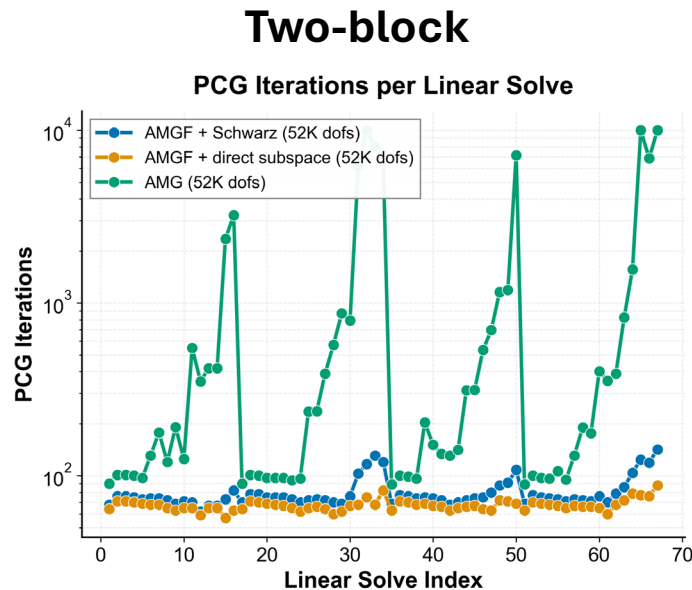
Red: nearly violated ($D > D_{min}$)
 Blue: not nearly violated ($D < D_{min}$)



Results

Convergence is between AMG and AMGF

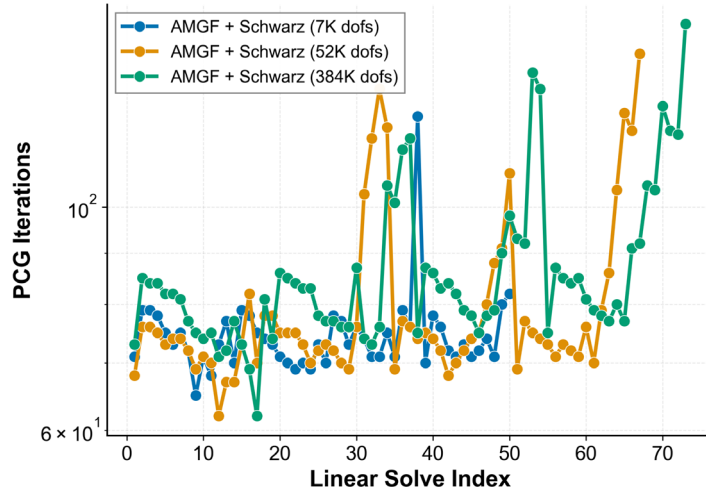
- For the two-block and ironing problems, AMGF-Schwarz exhibits comparable convergence behavior to AMGF.
- The beam-sphere problem is more challenging, though convergence is still far better than pure AMG.



Different problems behave differently under refinement

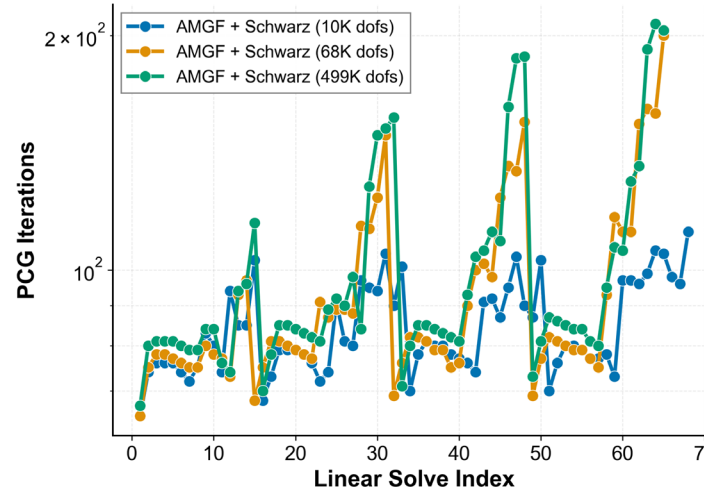
Two-block

PCG Iterations per Linear Solve



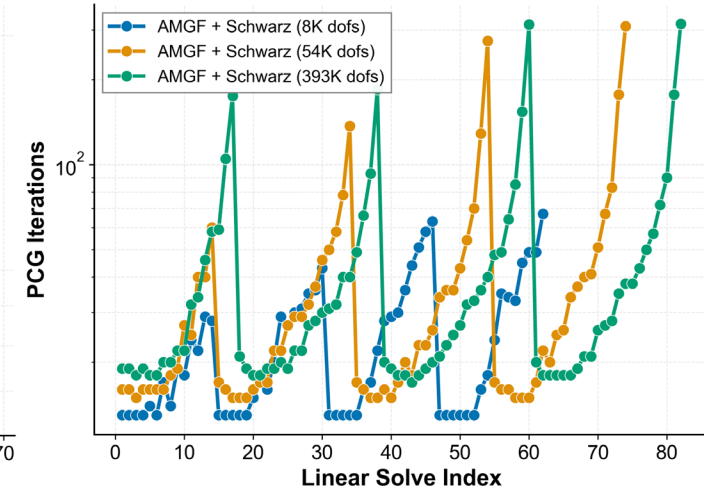
Ironing

PCG Iterations per Linear Solve



Beam-sphere

PCG Iterations per Linear Solve



DoFs (mean PCG iterations)

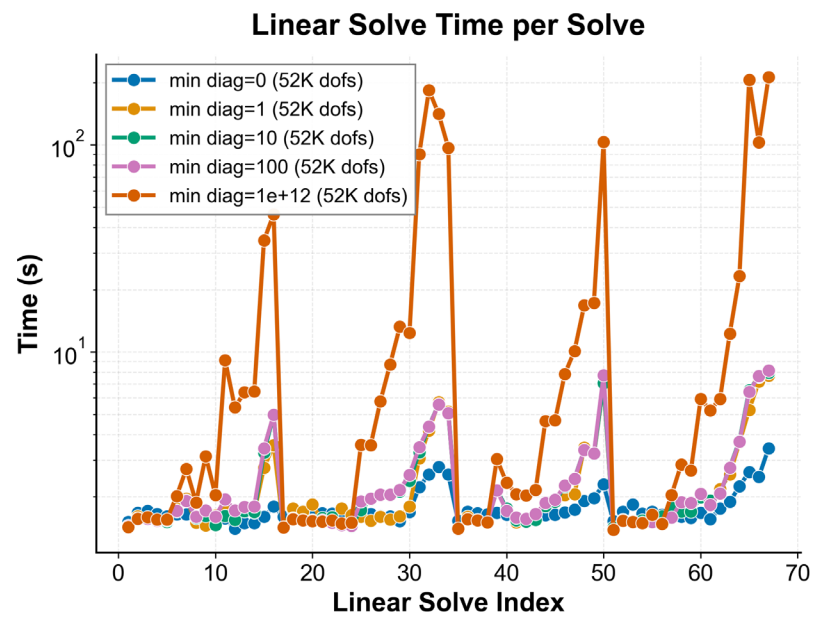
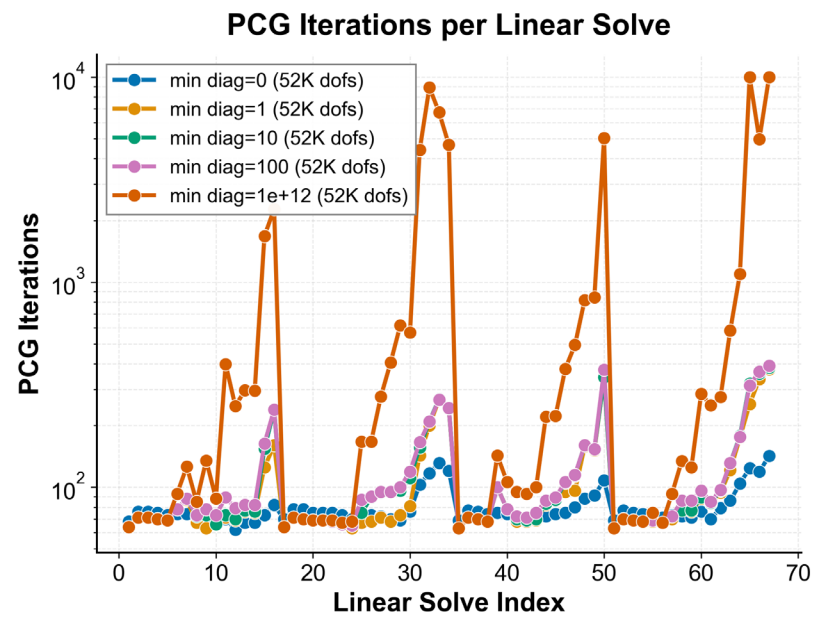
Refinement Levels	Two-block	Ironing	Beam-sphere
0	7K (75)	10K (84)	8K (25)
1	52K (80)	66K (94)	54K (40)
2	384K (87)	499K (101)	393K (44)

Only nearly violated constraints require correction

- Significant convergence benefit can be attained with only a few select Schwarz subdomains.

Two-block

$D_{\min}$	Subdomain Selection
0	100%
1	19%
10	13%
100	10%
inf	0%



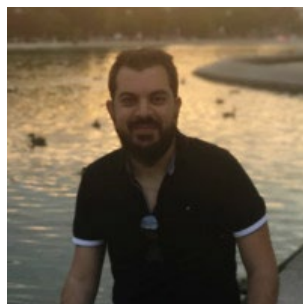
Thank you!

- We present an additive Schwarz-style subspace correction based on contact constraints that yields comparable convergence to AMG on three model problems.
- Future work includes timing studies, contact-rich problems, improved subdomain filtering, and development of theory.

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Questions?

Sensitivity to w

